



Topic 6: Bandwidth Utilization

CSE 4405: Data and Telecommunications

Department of CSE

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Definition

Bandwidth utilization is the wise use of available bandwidth to achieve communication goals.

- **Efficiency goal:** serve more traffic on the same infrastructure.
- **Security/robustness goal:** reduce eavesdropping and jamming impact.
- In this chapter, these goals are addressed through **multiplexing** and **spread spectrum**.

6.1 Multiplexing: Scope

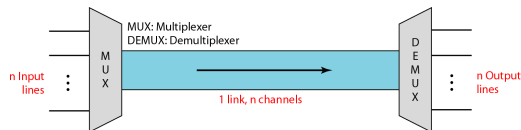
- Multiplexing allows simultaneous transmission of multiple signals over one shared link.
- It is used when link bandwidth is larger than each individual source requirement.
- As communication demand grows, multiplexing becomes essential for cost-effective scaling.

Topics in this section

- Frequency-Division Multiplexing (FDM)
- Wavelength-Division Multiplexing (WDM)
- Synchronous Time-Division Multiplexing (TDM)
- Statistical Time-Division Multiplexing

Multiplexing Basics: Link vs Channel

- **Link:** the physical path (cable, fiber, radio path).
- **Channel:** a logical portion of the link carrying one transmission stream.
- Multiplexing divides one physical link into multiple channels.



Multiplexing and Demultiplexing

Multiplexing

Combine several input streams into one composite stream (*many-to-one*).

Demultiplexing

Split the composite stream back into its component streams (*one-to-many*) and deliver each to the intended receiver.

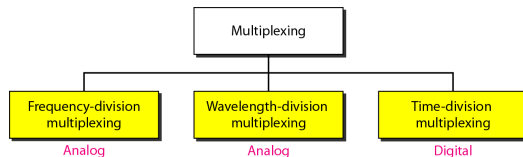
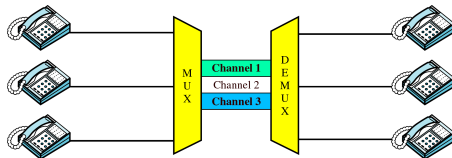


Figure: The shared link carries a combined signal between a MUX and a DEMUX.

Frequency-Division Multiplexing (FDM)

FDM is an **analog multiplexing** technique that combines analog signals by assigning each one a separate carrier frequency band.

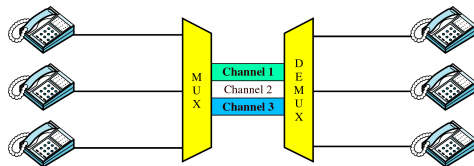
- Each source occupies a different frequency slot.
- All slots are transmitted simultaneously over the same link.
- Receiver-side filters separate the bands before demodulation.



FDM Process: Multiplexing Side

- Each telephone/data source generates a baseband signal.
- Each signal modulates a different carrier ($f_1, f_2, f_3, \dots$).
- Modulated signals are combined into one composite spectrum.

Result: one shared analog link carries multiple conversations simultaneously.



FDM Multiplexing in the Frequency Domain

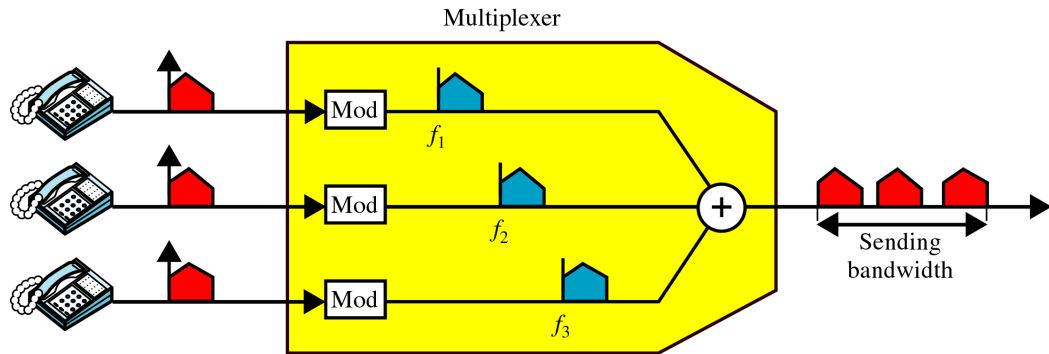
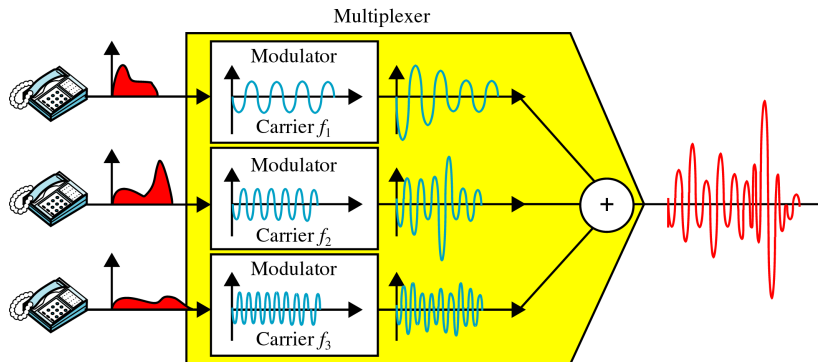


Figure: Separate subbands are allocated side-by-side across frequency, then combined onto one link.

FDM Multiplexing in the Time Domain

Figure: Although users share one medium, each keeps a dedicated frequency band continuously over time.



FDM Process: Demultiplexing Side

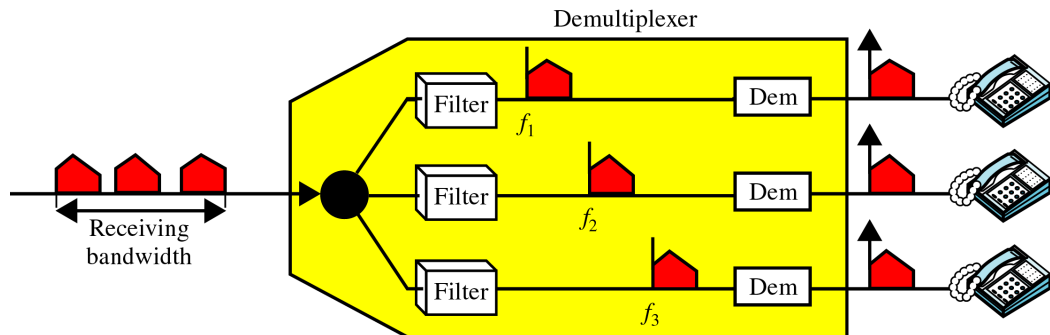
- At the receiver, the composite signal enters a demultiplexer.
- Band-pass filters isolate each carrier band.
- Each isolated band is demodulated to recover the original signal.
- Recovered streams are forwarded to the corresponding destinations.

Key point

Demultiplexing reverses multiplexing by frequency-selective separation.

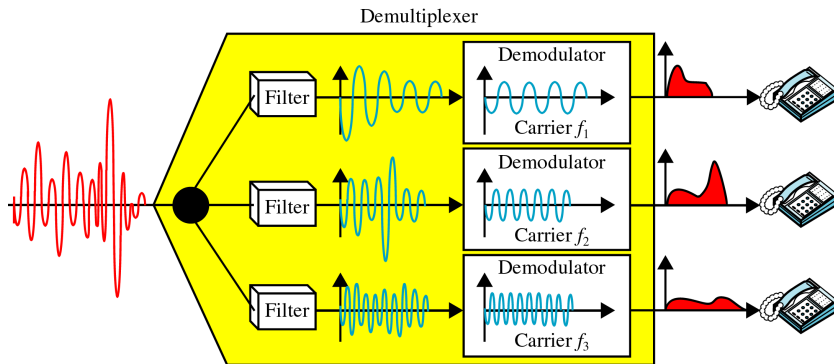
FDM Demultiplexing in the Frequency Domain

Figure: The composite spectrum is split back into original channel bands before delivery.



FDM Demultiplexing in the Time Domain

Figure: Each destination receives only its own continuous signal after filtering and demodulation.



Example 6.1: Three Voice Channels

Problem

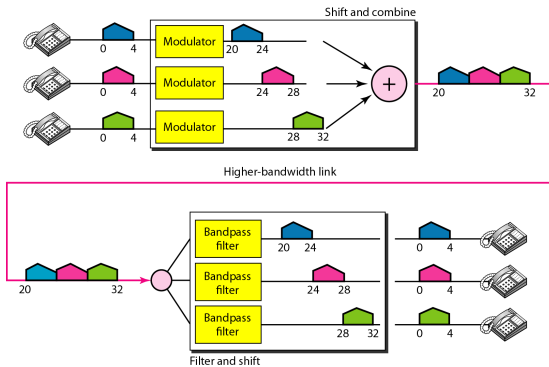
- One voice channel bandwidth: 4 kHz
- Combine 3 channels into 20 to 32 kHz (12 kHz total)
- Assume no guard bands

Solution idea

- Shift channels to 20–24, 24–28, and 28–32 kHz
- Then combine the three modulated channels into one link spectrum

Figure 6.6: Example 6.1

Figure: Three 4 kHz channels are translated to adjacent 4 kHz bands and combined.



Example 6.2: Guard Bands

Problem

- 5 channels, each 100 kHz
- Guard band required between adjacent channels: 10 kHz

Minimum link bandwidth

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz}$$

Why 4 guard bands? For N channels in one block, there are $N - 1$ boundaries between adjacent channels.

Figure 6.7: Example 6.2

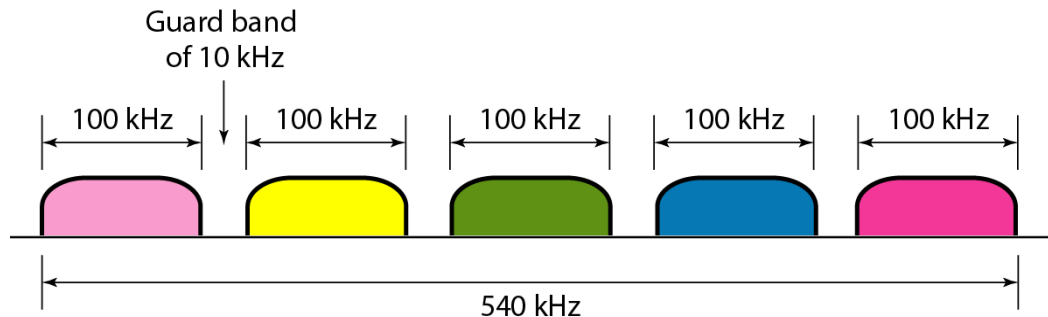


Figure: Guard bands are inserted between channel bands to reduce adjacent-channel interference.

Example 6.3: Digital Sources over an Analog Satellite Channel

Problem

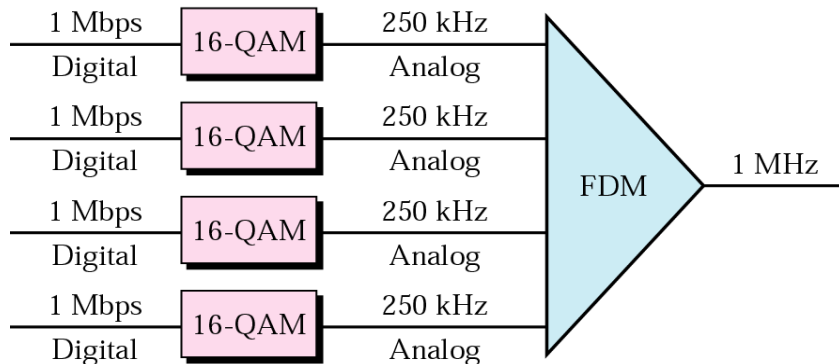
- Four digital channels, each 1 Mbps
- One analog satellite channel, total bandwidth 1 MHz

Design idea

- Split satellite bandwidth into four 250 kHz subchannels
- Modulate each digital source to fit its subchannel
- One possible modulation choice: 16-QAM (4 bits/symbol)

Figure 6.8: Example 6.3

Figure: Digital streams are modulated, frequency-separated, and carried together over one analog satellite band.



FDM Application: Cable Television

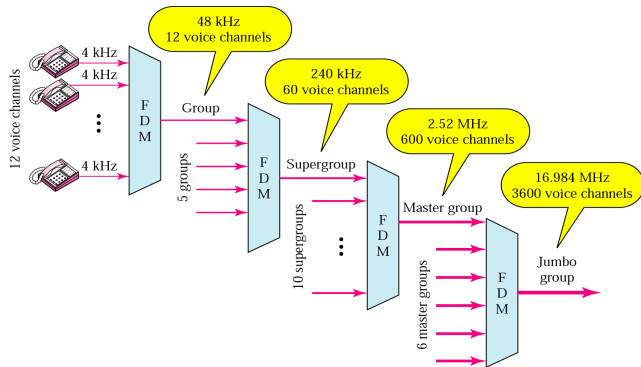
- Typical coaxial cable bandwidth: approximately 500 MHz
- One TV channel needs about 6 MHz
- Theoretical capacity: $\lfloor 500/6 \rfloor \approx 83$ channels

Engineering note

Practical channel count is lower after accounting for guard bands, control channels, and spectrum planning constraints.

Analog Hierarchy in Telephony

Figure: Telephone infrastructure multiplexing hierarchy



Example 6.4: AMPS Capacity

Given

- Uplink band: 824–849 MHz, downlink band: 869–894 MHz
- Per-user bandwidth per direction: 30 kHz

Capacity estimate

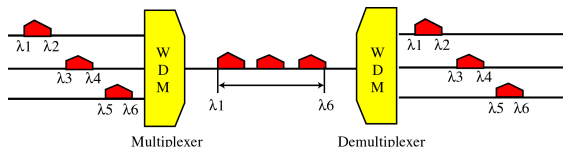
- Bandwidth per band: 25 MHz $\Rightarrow 25,000/30 \approx 833$ channels
- Practical allocation used 832 channels
- 42 channels reserved for control $\Rightarrow 790$ user channels

WDM: Optical Counterpart of FDM

WDM idea

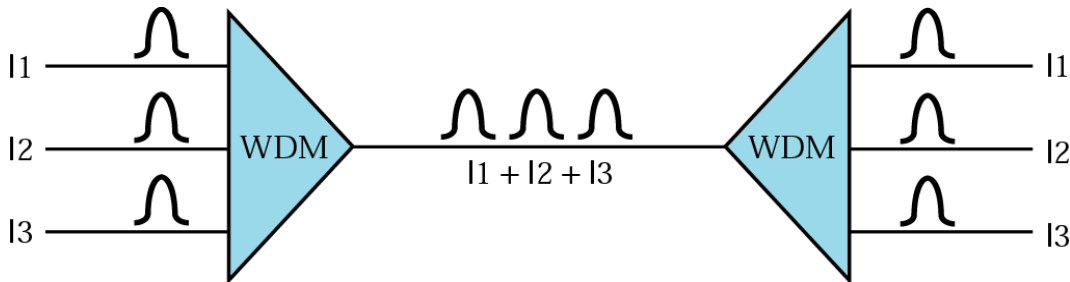
WDM is conceptually the same as FDM, but in fiber optics: channels are separated by **wavelength** instead of radio frequency bands.

- Multiple optical carriers ($\lambda_1, \lambda_2, \lambda_3, \dots$) share one fiber.
- Optical MUX combines wavelengths; optical DEMUX separates them.
- Used to exploit the very high bandwidth of fiber links.



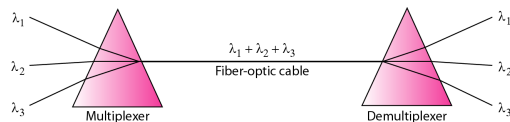
WDM in One Picture

Figure: Different light wavelengths coexist on the same fiber, each carrying an independent data channel.



Prism Intuition for WDM

- A prism bends light by wavelength-dependent refraction.
- The same physical principle explains optical combining and splitting.
- Optical transport systems such as SONET/SDH can use WDM layers for capacity growth.



Time-Division Multiplexing (TDM)

TDM is a **digital multiplexing** process in which multiple low-rate channels share one high-rate link by taking turns in time.

- Use TDM when link capacity exceeds per-source data rate.
- Time is divided into slots; slots are grouped into frames.
- Each source transmits in its assigned slot(s).

TDM Schemes

- **Synchronous TDM:** fixed slot assignment in every frame.
- **Statistical TDM:** slots assigned dynamically to active inputs.

Figure: Both schemes share a high-bandwidth link in time; they differ in slot assignment policy.

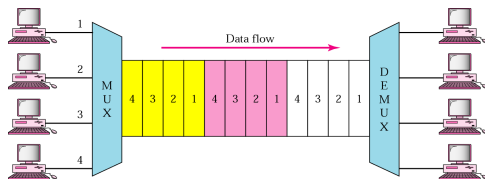
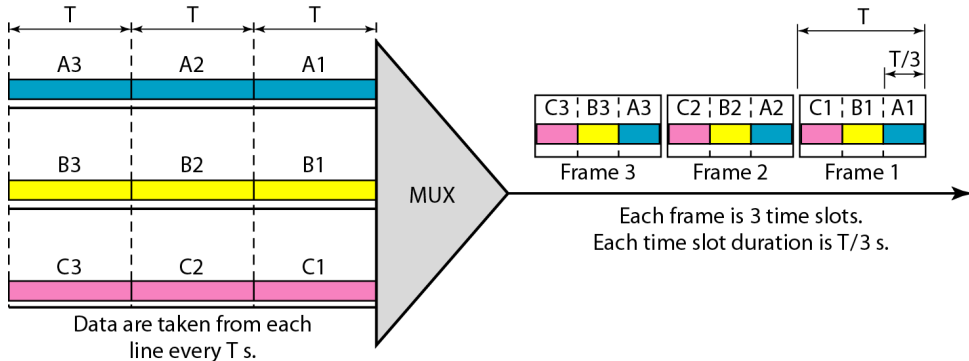


Figure 6.13: Synchronous TDM

Figure: In synchronous TDM with n inputs, link rate is n times faster and each output slot is $1/n$ of input-unit duration.



Example 6.5: Timing from Figure 6.13

Given each input rate is 1 kbps, and one bit is multiplexed per slot.

1. Input bit duration = $1/1000 \text{ s} = 1 \text{ ms}$
2. Output slot duration = $1/3 \text{ ms}$ (three inputs)
3. Frame duration = $3 \times (1/3 \text{ ms}) = 1 \text{ ms}$

Takeaway: frame duration equals the input unit duration when one unit per source is interleaved per frame.

Example 6.6: Bit and Frame Rates

Given Figure 6.14 has four 1 Mbps inputs, one-bit unit.

1. Input bit duration: $1/1 \text{ Mbps} = 1 \mu\text{s}$
2. Output bit duration: $1/4 \mu\text{s}$
3. Output bit rate: 4 Mbps
4. Output frame rate: 1,000,000 frames/s

Check: $\text{frame rate} \times \text{bits/frame} = 10^6 \times 4 = 4 \times 10^6 \text{ bps}$.

Figure 6.14: Synchronous TDM (Example 6.6)

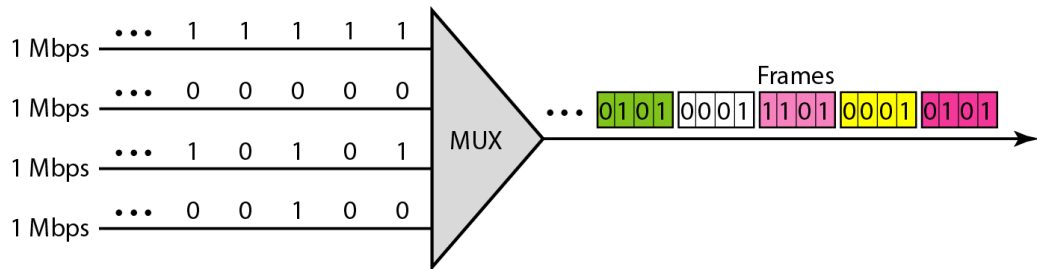


Figure: One bit from each input is interleaved into each frame, creating a higher-rate output stream.

Example 6.7: Four 1 kbps Inputs

Given four 1 kbps connections, unit = 1 bit.

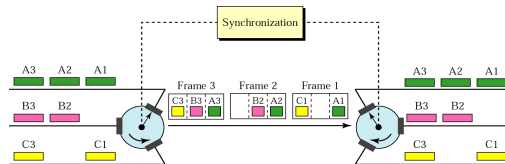
1. Input bit duration: $1/1 \text{ kbps} = 1 \text{ ms}$
2. Link rate: $4 \times 1 \text{ kbps} = 4 \text{ kbps}$
3. Slot duration: $1/4 \text{ ms} = 250 \mu\text{s}$
4. Frame duration: $4 \times 250 \mu\text{s} = 1 \text{ ms}$

Synchronous TDM: Interleaving

- Interleaving can be visualized as a very fast rotating switch.
- On the MUX side, each input sends when its slot position arrives.
- On the DEMUX side, the receiving switch distributes units in the same order.
- Correct operation requires synchronized slot timing.

Synchronous TDM: Empty Slots

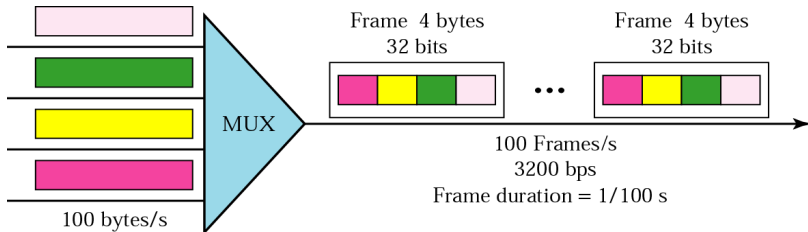
- Fixed slot assignment is simple and predictable.
- If a source is idle, its slot in that frame becomes empty.
- Empty slots reduce bandwidth efficiency for bursty traffic.



Example 6.8: Byte-Level Multiplexing

Given 4 channels, each 100 bytes/s, multiplex 1 byte per channel per frame.

- Frame size: $4 \times 1 \text{ byte} = 4 \text{ bytes} = 32 \text{ bits}$
- Frame rate: 100 frames/s (to serve each channel's 100 bytes/s)
- Frame duration: $1/100 = 0.01 \text{ s} = 10 \text{ ms}$
- Link bit rate: $100 \times 32 = 3200 \text{ bps} = 3.2 \text{ kbps}$



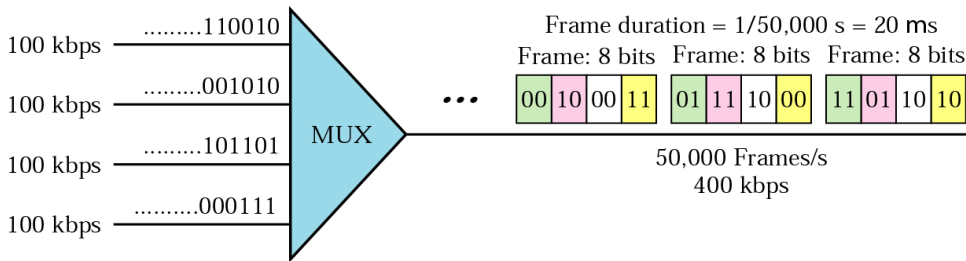
Example 6.9: 2-Bit Slots

Given four 100 kbps channels, time slot = 2 bits/channel.

- Frame carries $4 \times 2 = 8$ bits.
- Frame rate: $100,000/2 = 50,000$ frames/s
- Frame duration: $1/50,000 = 20 \mu s$
- Link bit rate: $50,000 \times 8 = 400$ kbps
- Bit duration on link: $1/400,000 = 2.5 \mu s$

Example 6.9 Output Illustration

Figure: Arbitrary 2-bit chunks from each channel are interleaved into 8-bit frames at 50,000 frames/s.

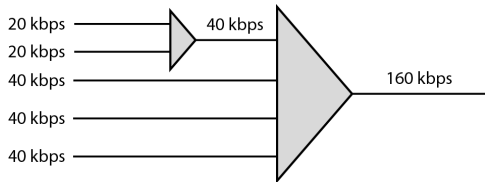


Synchronous TDM: Data-Rate Management

- Real systems often have unequal input rates.
- Three standard strategies are used to handle rate disparity:
 - Multilevel multiplexing
 - Multiple-slot allocation
 - Pulse stuffing (bit padding)

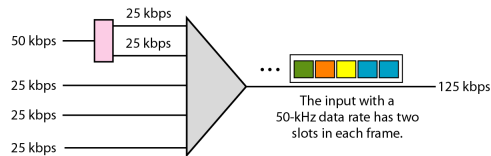
Multilevel Multiplexing

- Used when one input rate is an integer multiple of another.
- First, combine lower-rate inputs into an intermediate stream.
- Then perform higher-level multiplexing with other streams.
- This creates a clean hierarchical design.



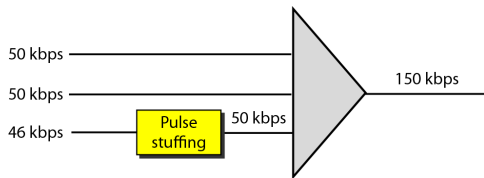
Multiple-Slot Allocation

- Give faster sources more than one slot per frame.
- A serial-to-parallel converter can split one high-rate input into multiple logical slot streams.
- Example: a 50 kbps line may be assigned two slots while lower-rate lines get one.



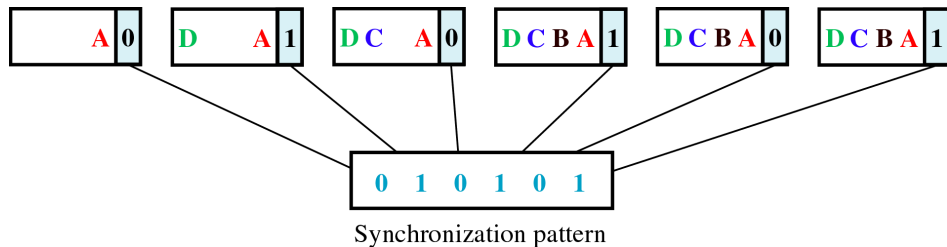
Pulse Stuffing (Bit Padding)

- Used when rates are not integer multiples.
- Choose the highest source rate as the reference.
- Add dummy bits to lower-rate streams to match timing.
- Also called **bit padding** or **bit stuffing**.



Framing Synchronization

- If MUX and DEMUX lose frame alignment, bits can be delivered to wrong channels.
- Framing bits are inserted so the receiver can detect slot boundaries.
- A common pattern is a 1-bit sync stream (e.g., alternating 101010...).
- Synchronization bits are typically placed at a fixed frame position.



Example 6.10: Character Streams with Framing Bit

Given 4 sources, each 250 characters/s, 8 bits/character, plus 1 framing bit/frame.

1. Source rate: $250 \times 8 = 2000 \text{ bps} = 2 \text{ kbps}$
2. Character duration: $1/250 = 4 \text{ ms}$
3. Frame rate: 250 frames/s
4. Frame duration: 4 ms
5. Bits/frame: $4 \times 8 + 1 = 33$
6. Link rate: $250 \times 33 = 8250 \text{ bps}$

Example 6.11: Unequal Input Rates

Problem Multiplex 100 kbps and 200 kbps channels. **Design**

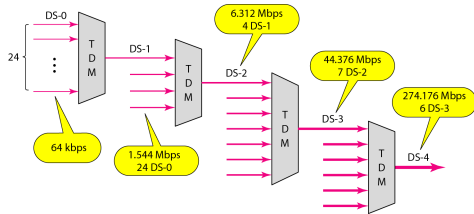
- Allocate 1 slot/frame to 100 kbps channel.
- Allocate 2 slots/frame to 200 kbps channel.
- So each frame carries 3 bits total.

Results

- Frame rate: 100,000 frames/s
- Frame duration: $10\ \mu\text{s}$
- Link bit rate: $100,000 \times 3 = 300\ \text{kbps}$

Digital Signal (DS) Service

- Digital hierarchy defines standard multiplexed carrier rates.
- Compared with analog trunks, digital services are generally:
 - less sensitive to noise
 - more cost-effective for large-scale deployment



DS Service Rate Family

Level	Rate	Composition intuition
DS-0	64 kbps	Single digital voice/data channel
DS-1	1.544 Mbps	24×64 kbps plus overhead
DS-2	6.312 Mbps	96×64 kbps plus overhead
DS-3	44.376 Mbps	672×64 kbps plus overhead
DS-4	274.176 Mbps	4032×64 kbps plus overhead

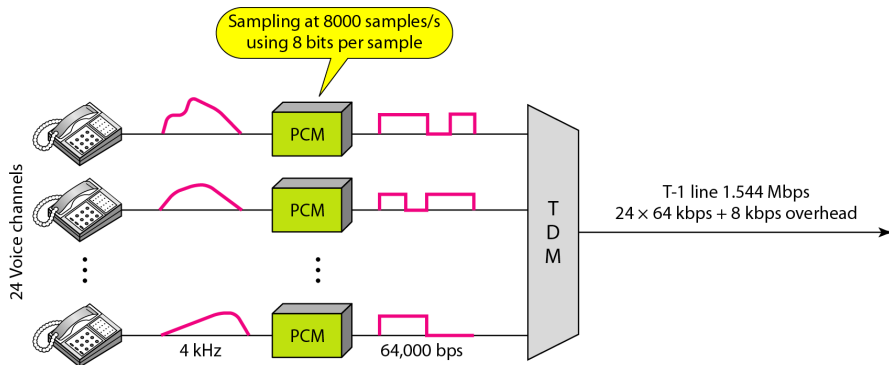
Table 6.1: DS and T-Line Rates

Figure: DS levels map to practical T-carrier transmission rates used in digital telephony infrastructure.

<i>Service</i>	<i>Line</i>	<i>Rate (Mbps)</i>	<i>Voice Channels</i>
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

Figure 6.24: T-1 for Telephone Multiplexing

Figure: A T-1 line multiplexes many voice channels into one framed digital stream for trunk transport.

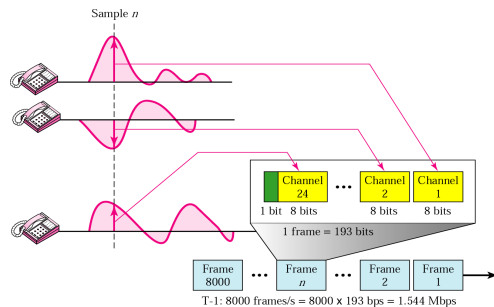


T-1 Frame Structure

Frame size

$$193 = 24 \times 8 + 1$$

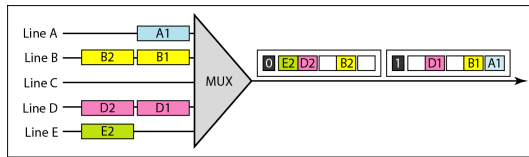
- 24 channels $\times$ 8 bits = 192 payload bits
- 1 additional framing bit for synchronization



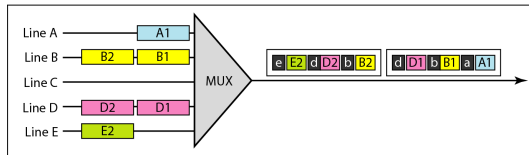
Statistical TDM: Motivation

- Slots are assigned dynamically to active sources.
- In synchronous TDM, fixed slots can be wasted when lines are idle.
- In statistical TDM, no slot is left empty while any input has data waiting.
- This improves bandwidth efficiency for bursty traffic.

Figure 6.26: Slot Comparison



a. Synchronous TDM



b. Statistical TDM

Statistical TDM: Addressing and Synchronization

- Because slot positions vary, each slot carries **address information**.
- If N output lines exist, n address bits are needed where:

$$n = \log_2 N$$

- Frame-wide synchronization bits used in synchronous TDM are typically unnecessary in the same form.
- The receiver uses slot headers to deliver payloads correctly.

6.2 Spread Spectrum: Scope

- Spread spectrum (SS) intentionally uses bandwidth wider than the minimum required for raw data.
- Main goals are **anti-jamming robustness** and **reduced interception vulnerability**.
- Redundancy is introduced during spreading to gain these properties.

Techniques in this section

- Frequency-Hopping Spread Spectrum (FHSS)
- Direct-Sequence Spread Spectrum (DSSS)

Figure 6.27: Spread Spectrum Principles

- Allocate more bandwidth than the original data stream strictly needs.
- Perform spreading after source signal generation.
- Receiver performs inverse operation (despreading) using matching strategy/code.

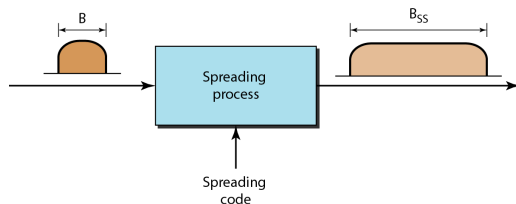


Figure 6.28: FHSS

Figure: In FHSS, a signal hops among multiple carrier frequencies over time according to a hopping sequence.

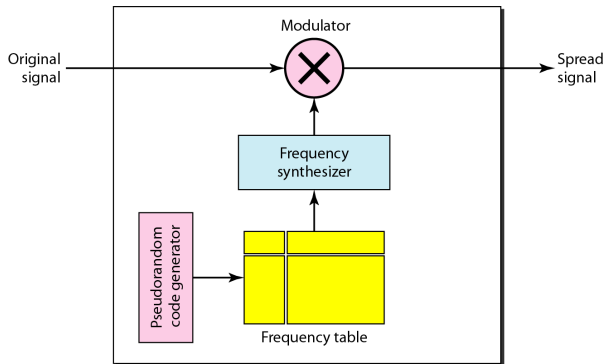


Figure 6.29: Frequency Selection in FHSS

Figure: A pseudo-random control process selects the next carrier frequency for each hop interval.

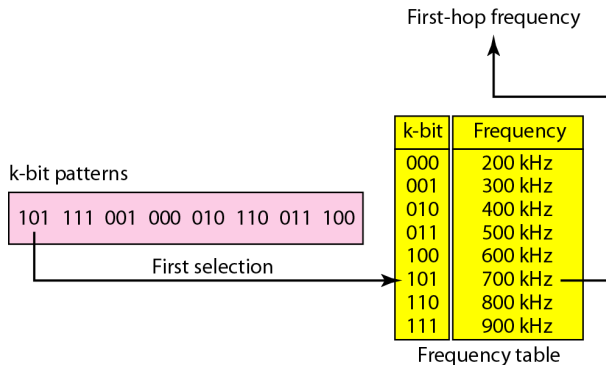


Figure 6.30: FHSS Cycles

Figure: Hopping patterns repeat in cycles; transmitter and receiver must stay sequence-synchronized.

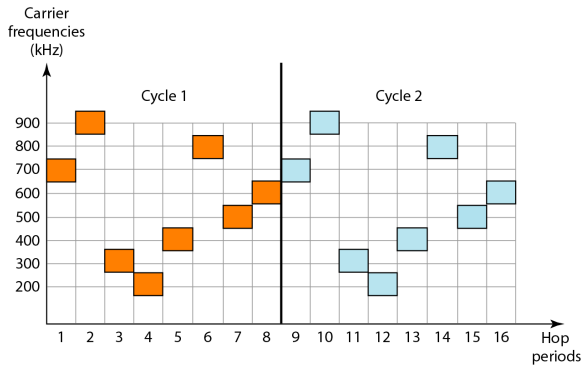
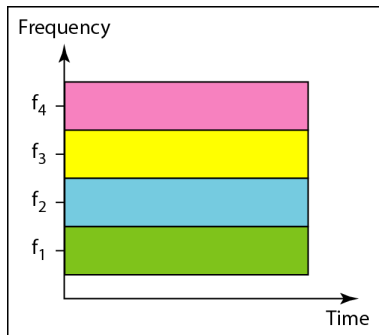
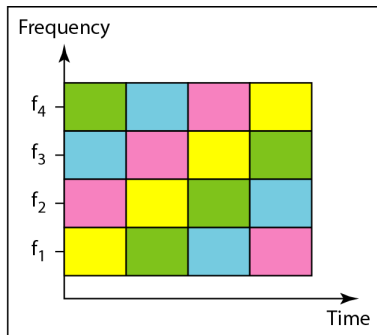


Figure 6.31: Bandwidth Sharing in FHSS

Figure: With M hopping frequencies, multiple channels can share the spread band through coordinated hopping patterns.



a. FDM



b. FHSS

Figure 6.32: DSSS

- Each data bit is replaced by n faster chips.
- Chips come from a spreading code.
- Chip rate is n times the data-bit rate.

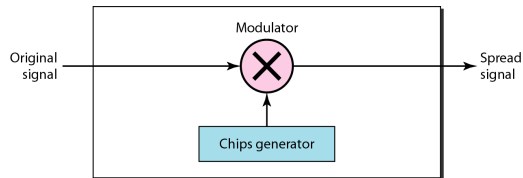
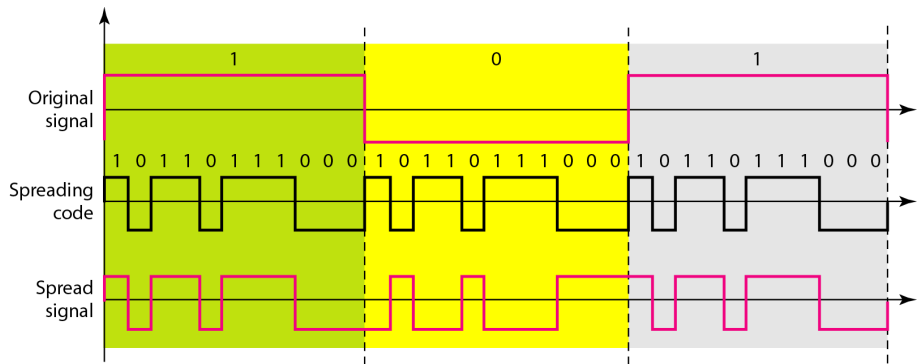


Figure 6.33: DSSS Example

Figure: Bit-by-chip spreading expands occupied bandwidth, enabling later despreading with the correct code.



Summary (1)

- Bandwidth utilization means using available bandwidth to achieve specific goals.
- **Multiplexing** improves efficiency; **spreading** improves robustness/privacy against jamming and interception.
- Multiplexing enables simultaneous transmission of multiple signals over one data link.
- Three core multiplexing families are FDM, WDM, and TDM.
- WDM applies analog multiplexing ideas to optical wavelengths in fiber systems.

Summary (2)

- TDM is a digital process that shares one high-rate link among several low-rate streams in time.
- Synchronous TDM uses fixed slot positions even when a source is idle.
- Statistical TDM assigns slots dynamically and improves efficiency for bursty traffic.
- Spread spectrum is especially useful in wireless settings where coexistence, anti-jamming, and anti-interception matter.

Summary (3)

- FHSS spreads by hopping carriers among many frequencies over time.
- DSSS spreads by replacing each data bit with a chip sequence.
- Both techniques trade extra bandwidth for improved robustness and sharing behavior in hostile or crowded spectrum.